

# Jun Woo Beck

Audio & Music Machine Learning Researcher · Music Informatics Group, Georgia Tech

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## RESEARCH SUMMARY

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MS student in Music Technology at Georgia Tech's Music Informatics Group, researching machine learning and music information retrieval for sound effects and music audio. **First author of a DAFx 2026 paper** on sound-effects dataset unification. Brings professional musicianship and audio engineering experience to audio ML research.

## RESEARCH INTERESTS

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Multimodal video–music understanding · music information retrieval (MIR) · audio–visual representation learning · sound effects & music-audio machine learning

## EDUCATION

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### Georgia Institute of Technology

*Master of Science in Music Technology*

Music Informatics Group · Advisor: Dr. Alexander Lerch

Relevant coursework: Machine Learning, Audio Content Analysis, Integrating Music into Multimedia

Atlanta, GA

Aug 2025 – Expected May 2027

### Berklee College of Music

*Bachelor of Music in Electronic Production and Design*

Minors: Game and Interactive Media Scoring; Screen Scoring

**Magna Cum Laude** · GPA 3.71 / 4.00

Relevant coursework: Digital Signal Processing for Music Production and Postproduction, Music Acoustics

Boston, MA

Aug 2024

## PUBLICATIONS

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*Peer-Reviewed Conference Proceedings*

**Beck, J. W.**, & Lerch, A. (2026). *Sound Effects Dataset Unification With the Universal Category System*. In Proc. 29th International Conference on Digital Audio Effects (DAFx26), Cambridge, MA, USA, pp. 413–420.

Oh, E. J., **Beck, J. W.**, & Smith, A. (2026). *The Singing Skin: An Audience-Centered Biofeedback System for Musical Interaction Based on Galvanic Skin Response*. In Proc. 51st International Computer Music Conference (ICMC), Hamburg, Germany, pp. 255–260.

*Extended Abstracts (Late-Breaking / Demo)*

Oh, E. J., Park, S., Kim, Y., **Beck, J. W.** (2026). *A User-Centered View of AI Music Description*. Late-Breaking/Demo, 27th International Society for Music Information Retrieval Conference (ISMIR 2026), Abu Dhabi, UAE. [accepted]

## RESEARCH EXPERIENCE

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### Music Informatics Group, Georgia Institute of Technology

*Graduate Researcher — Advisor: Dr. Alexander Lerch*

Research spanning sound-effects dataset unification (**DAFx 2026**) and an industry-sponsored acoustic evaluation with **Textron**; ongoing research on multimodal video–music understanding for film scoring.

Atlanta, GA

Aug 2025 – Present

## SELECTED PROJECTS

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### Sound Effects Dataset Unification — EnvSound-UCS (DAFx 2026)

Fall 2025 – Spring 2026

- Built a rule-based tag-to-UCS conversion framework reaching 98.5–100% automatic conversion across FSD50K, AudioSet, and ESC-50.
- Released EnvSound-UCS, a unified 58,057-clip environmental-sound corpus spanning 59 categories, and validated it with audio-classification benchmarks.
- Open-sourced the conversion pipeline, tools, and converted datasets across four public repositories ([github.com/JunWooBeck](https://github.com/JunWooBeck)).

### Bio Sing-Along — Interactive Biosignal Performance

Spring 2026

Premiered May 2026 at PRISM Pt. 1: Sound Through New Lens, Georgia Tech Center for Music Technology · with Eun Ji Oh

- Built the audio system — sound morphing and real-time processing — converting live GSR and PPG biosignals into the performance's sonic material.

### Textron Panel Project — Acoustic Evaluation & Digital EQ

Spring 2026

Industry collaboration with Textron · Music Informatics Group, Georgia Tech

- Conducted controlled acoustic measurements (THD, SINAD, frequency response) of speaker-panel audio quality in an isolation room against a consumer-laptop reference.
- Designed parametric EQ tuning presets that substantially reduced measured harmonic distortion and flattened the frequency response.

### The Singing Skin — Audience-Centered Biofeedback System (ICMC 2026)

Fall 2025

- Designed and implemented the full audio system translating real-time galvanic skin response (GSR) signals into interactive musical output.

## HONORS & AWARDS

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GTCMT Seed Grant (\$2,000) — Georgia Tech Center for Music Technology

2026

Magna Cum Laude — Berklee College of Music

2024

Dean's List (4 semesters) — Berklee College of Music

2020 – 2023

## SKILLS

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**Programming:** Python, Max/MSP

**Machine Learning:** model training & evaluation; audio dataset construction & curation

**Audio Engineering:** recording & mixing (Pro Tools, Logic Pro, iZotope RX)

**Sound Design:** synthesis and sound-effects design for games, film, and video (FMOD, Wwise)

**Music:** composition, string arranging & orchestration; violin performance (classical & jazz)

## MUSIC PRODUCTION & PERFORMANCE

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### Recording / Mixing / Live Sound Engineer — Seoul

2021

Mixed and mastered orchestral music with traditional Korean instruments (**Korean Ministry of Culture, Sports and Tourism**, dir. Echaé Kang); recorded and mixed a full string-orchestra performance and designed live sound effects for dance (Seoul Ballet Festival, **Seoul Arts Center**).

### Violinist & String Arranger — Dear Jazz Orchestra · Other Orchestras

2017 – 2022

**Soloist** and orchestra member at the **Seoul Jazz Festival** and **Jarasum Jazz Festival**; recorded on multiple Dear Jazz Orchestra releases and arranged strings for "Sing for You Vol. 1"; performed with Berklee World Strings, Berklee Motion Picture Orchestra, and film-scoring session orchestras.